PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH
BADJI MOKHTAR UNIVERSITY -ANNABA-
FACULTY OF MEDICINE
DEPARTMENT OF MEDICINE
LABORATORY OF MEDICAL AND SURGICAL ANATOMY

Hip joint

Course notes intended for first-year medical students.

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Hip joint

Course outline

Introduction
1. Articular surfaces
1.1. Acetabulum
1.2. Acetabular Labrum
1.3. Femoral head
2. Means of union
2.1. Joint capsule
2.2. Passive ligaments
2.3. Ligament of the femoral head
2.4. Active ligaments (the periarticular muscles)
3. Synovial
4. Joint physiology
5. Clinical applications
Conclusion
Bibliography

Hip joint

Introduction

The hip joint, or coxofemoral joint, is the single joint that connects the lower limb to the trunk. It is a proximal ball-and-socket joint of the lower limb with a dual function: support and stability.
The hip supports the pelvis and orients the lower limb in all directions of space, ensuring the upright posture that characterizes humans. It is a deep, weight-bearing, mobile, and highly congruent joint.

1.Articular surfaces

1.1.The acetabulum

The acetabulum is a deep cavity located on the lateral surface of the hip bone, bordered by a prominent bony rim called the acetabular limbus. The acetabulum consists of two distinct parts:

- A central, non-articular part: the acetabular fossa.
- A peripheral, articular, crescent-shaped part: the semilunar surface.

1.2.The acetabular labrum

The acetabular labrum is a fibrocartilaginous ring attached the perimeter of the acetabulum; it passes over the ischiopubic notch, forming the transverse ligament of the acetabulum.
It equalizes the acetabular limbus and increases the depth and extent of the acetabulum.

1.3.The femoral head

The femoral head is a rounded articular projection, representing 2/3 of a sphere. At its center is the fovea capitis where the ligament of the femoral head (the ligamentum teres) attaches.

2. Means of union

The following maintain the joint surfaces in contact: the joint capsule, the passive ligaments, the ligament of the femoral head and the passive ligaments.

2.1.The joint capsule

The joint capsule is a fibrous sleeve which has an iliac insertion on the acetabular limbus and labrum and a femoral insertion on the intertrochanteric line and femoral neck.

2.2.Passive ligaments

2.2.1. Iliofemoral ligament (Bertin ligament)

The anterior ligament, triangular and strong, extends between the acetabulum and the intertrochanteric line. It has two distinct bundles :

- the superior bundle : ilio-pretrochanteric
- the inferior ilio-pretrochanteric bundle.

2.2.2. pubofemoral ligament

The pubofemoral ligament reinforces the anteroinferior part of the joint capsule and extends between the iliopectineal eminence and the pretrochanteric depression.

2.2.3. Ischiofemoral ligament

The ischiofemoral ligament reinforces the posterior joint capsule and extends between the ischium and femoral neck.

2.3. Ligament of the femoral head (round ligament)

The ligament of the femoral head is an intra-articular and extra-synovial ligament, centered on an artery that supplies the femoral head. It extends from the fovea capitis and terminates in three bundles:

- Anterior bundle: on the anterior horn of the semilunar surface.
- The middle bundle attaches to the transverse ligament.
- Posterior bundle on the ischium.

2.4.Active ligaments (periarticular muscles)

- Anterior : the Iliopsoas muscle, the rectus femoris muscle and the sartorius muscle.
- Lateral : the tensor fasciae latae muscle, the gluteus medius muscle, and the gluteus minimus muscle.
- Posterior : the gluteus maximus muscle, the pelvi-trochanteric muscles and the hamstring muscles.
- Medial : the pectineus muscle, the adductor longus muscle, the adductor magnus muscle

3. Synovial

The synovium lines the deep surface of the capsule.

The synovium of the ligament of the femoral head is separate and independent from the articular synovium; it attaches to the periphery of the fovea capitis and the acetabular fossa.

4. Joint physiology

The hip joint is a very stable joint with a wide range of motion. Its stability is ensured by the joint capsule, ligaments, and surrounding muscles. It allows:

- Flexion-extension movements
- Adduction-abduction movements
- Medial rotation – lateral rotation movements
- The circumduction

4.1.Flexion-extension movements

Flexion and extension occur around a transverse axis that passes through the center of the femoral head.

Flexion brings the anterior surface of the thigh closer to the anterior abdominal wall.
The range of motion: 120°: knee bent, 90°: knee extended.
The extension moves the anterior surface of the thigh away from the anterior abdominal wall
The range of motion: 10° with the knee bent and 20° with the knee extended

4.2.Adduction-abduction movements
Abduction and adduction occur around an anteroposterior axis that passes through the center of the femoral head.
Abduction spread the thigh away from the midline
The usual range of motion: 30°, forced: 45° to 60° (gymnastics…)
Adduction brings the thigh closer to the midline.
Maximum angle of rotation : 30°

4.3.The circumduction motion
Circumduction results from the succession of the 4 preceding movements.

4.4.The medial rotation – lateral rotation movements performed around a vertical axis that passes through the center of the femoral head, lateral rotation: 45°- 60°, medial rotation: 30°- 45°.

5. Clinical application

5.1.Radioanatomy
Numerous techniques are available today for analyzing the tendons, bone, capsuloligamentous structures, fibrocartilage, trophic state, and muscular content of the hip joint. These techniques utilize ultrasound, computed tomography (CT) scans, with or without arthrography, and magnetic resonance imaging (MRI), with or without arthrography.

5.2.Pathologies

- Osteoarthritis of the hip joint or coxarthrosis.
- Arthritis is a serious inflammation or infection (it can be treated with prostheses).
- Fractures of the femoral neck or acetabulum.
- Dislocations are rare because of the stability of the joint.
- Congenital hip dislocations (CHD) are detected at birth using the Ortolani maneuver (which assesses instability during hip mobilization). Orthopedic treatment involves placing the baby in a swaddling position with their hips in forced abduction.

Conclusion

The hip joint is a weight-bearing, ball-and-socket joint connecting the pelvis to the femur. It is both highly mobile and extremely stable thanks to its congruent structure and powerful connecting elements such as the capsule, ligaments, and muscles. Its anatomy allows for a wide range of motion essential for walking and posture, while simultaneously resolving the paradox of stability, making it one of the strongest joints in the human body.

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3. Rouvière H., Delmas A. Human Anatomy. Descriptive, Topographical and Functional. Volume 3. Limbs. Masson Edition, 2002, 667 pages.
4. Kamina P. Clinical Anatomy. Limbs. Maloine Edition, 1997, 287 Pages.
5.Ibrahim AdalbertKapandji. Joint Physiology. Volume 1: Upper Limb. Paris: Maloine 1980.

Articulation de la hanche

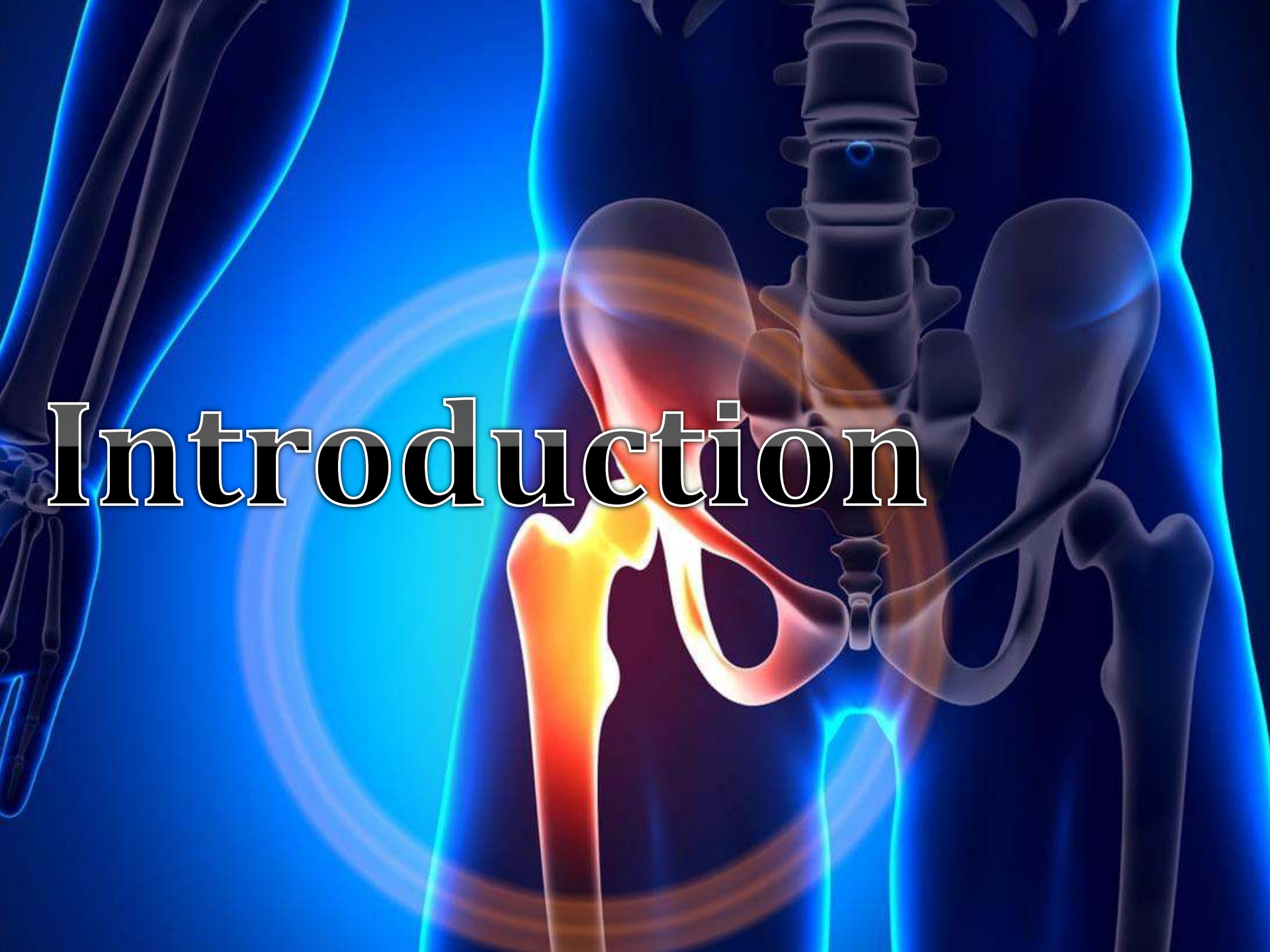


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Plan du cours

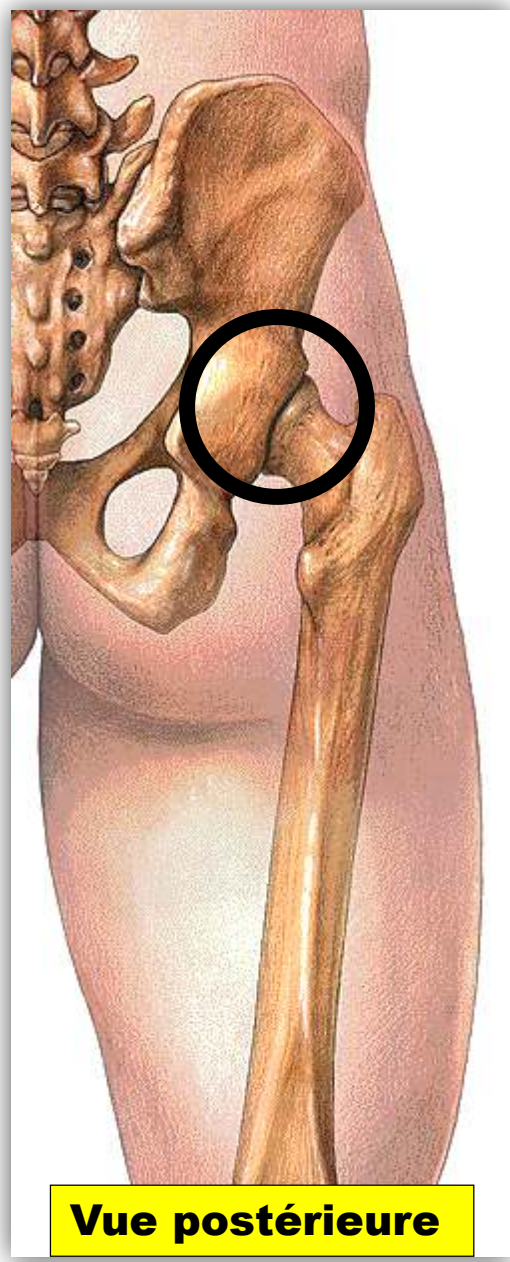
- Introduction
- Surfaces articulaires
- Moyens d'unions
- Synoviale
- Physiologie articulaire
- Anatomie clinique
  - Radio-anatomie
  - Pathologie



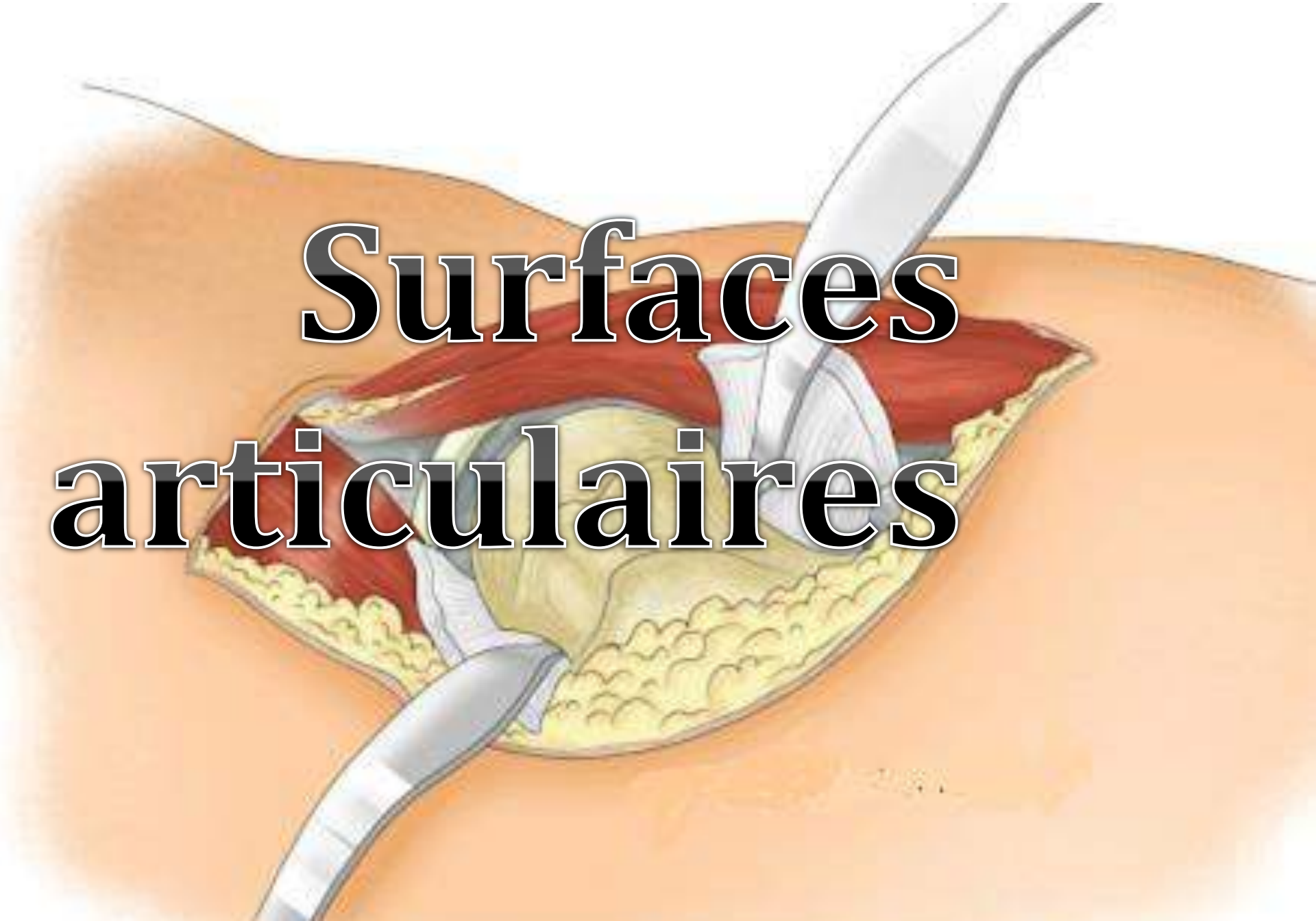

Introduction



- **L’articulation de la hanche ou coxo-fémorale, est l’articulation unique qui unie le membre inférieur au tronc.**
- **Articulation proximale** du membre pelvien de type **sphéroïde.**
- A double vocation pour le membre pelvien : **appuie** et **stabilité.**
- **La hanche supporte le bassin.**
- Elle oriente le membre inférieur dans toutes les directions de l'espace et assure la station **érigée** qui caractérise **l’homme.**
- Articulation **profonde**, **portante**, **mobile** et très **congruente.**



Surfaces articulaires

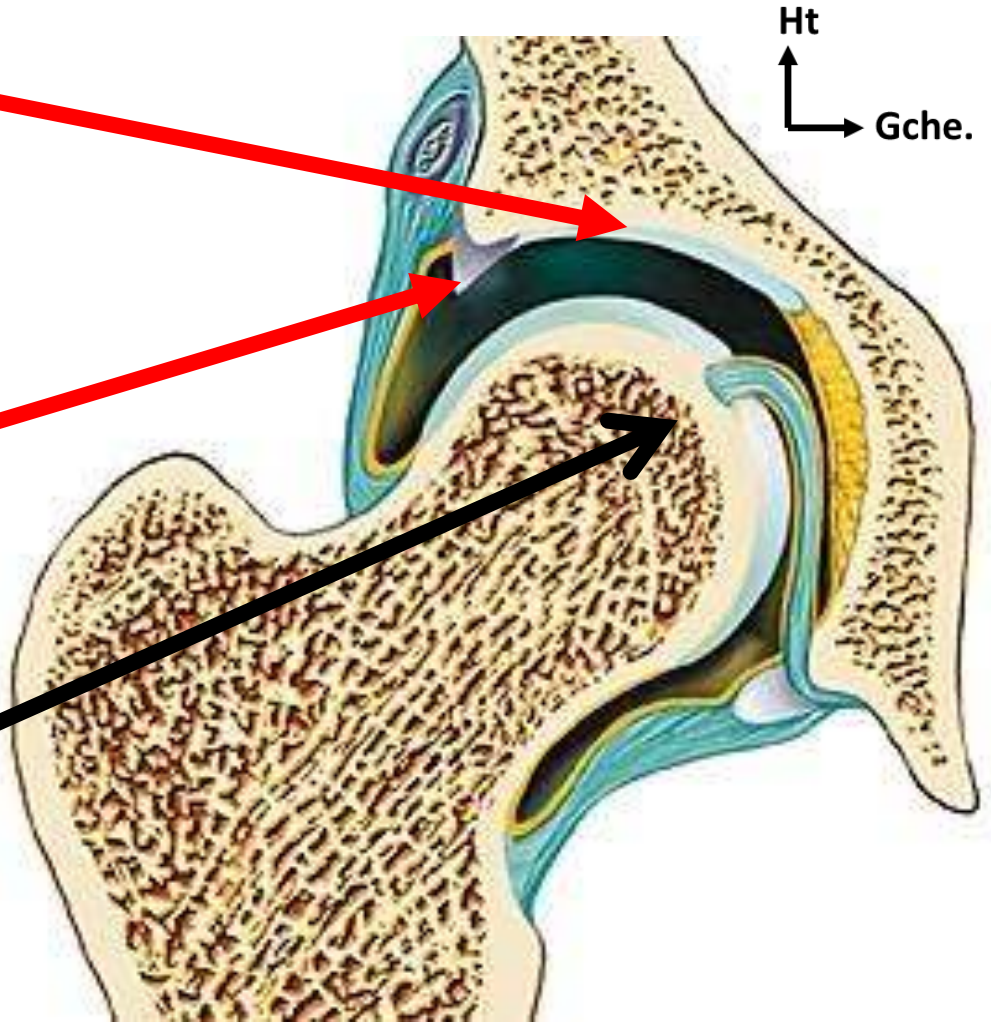


Représentées par:

L'acétabulum(cotyle)

Le labrum (bourrelet acétabulaire)

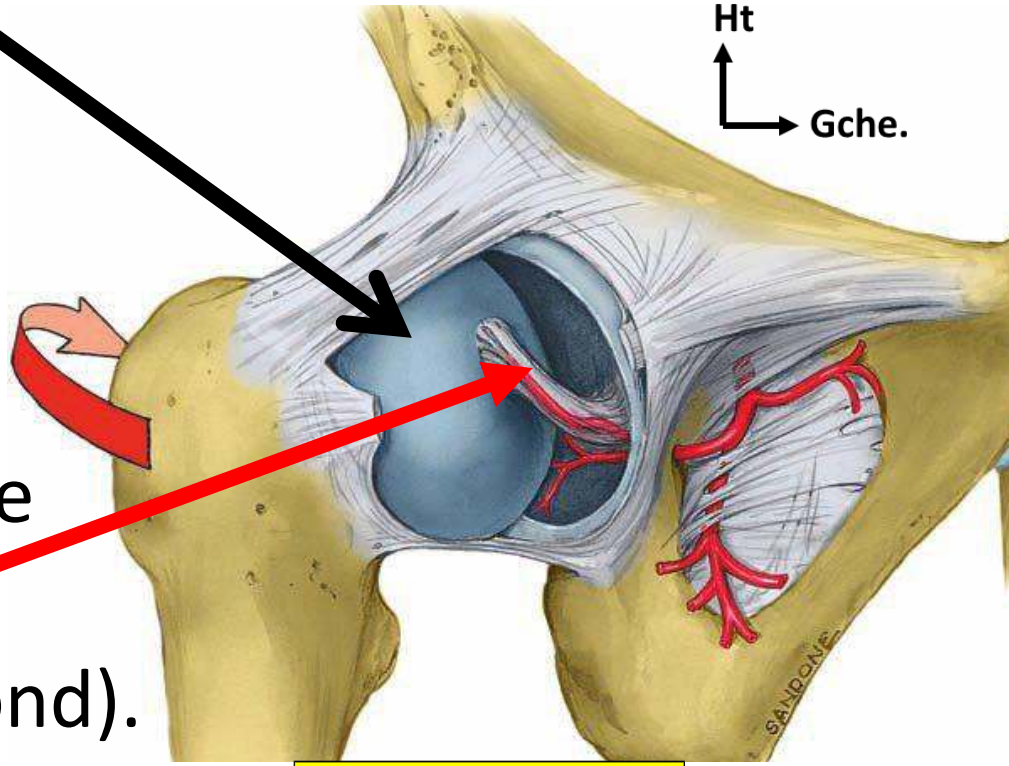
La tête fémorale



Coupe frontale

Tête fémorale

- **Une saillie articulaire arrondie**
- **2/3** d'une sphère
- Présente à son centre:
  la **fovéa capitis** où se fixe
  le **ligament de la tête fémorale** (le ligament rond).



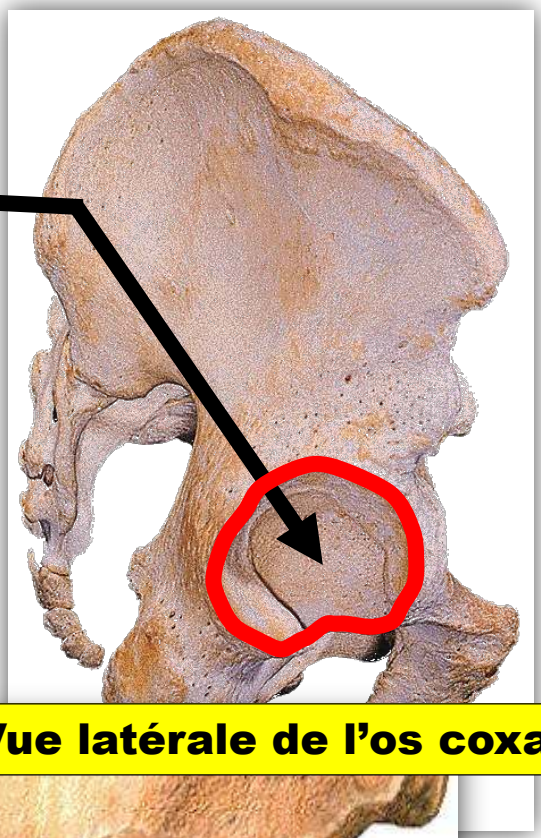
Vue antérieure

Acétabulum

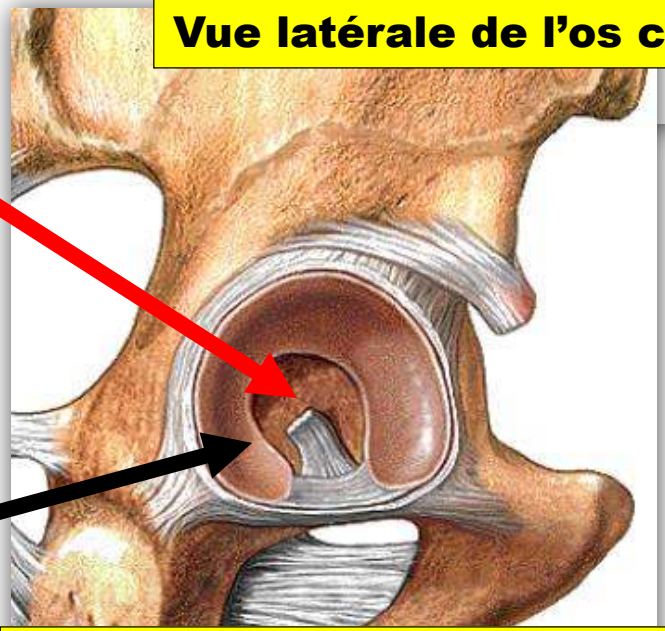
- **Une excavation profonde**
- Limitée par un rebord saillant: **le limbus acétabulaire**
- **Présente 2 parties** distinctes:
  - **Central, non articulaire**

    la **fosse de l'acétabulum**
  - **Périphérique, articulaire,** en croissant:

    la **surface semi-lunaire**



Vue latérale de l'os coxal



Vue latérale de l'os coxal

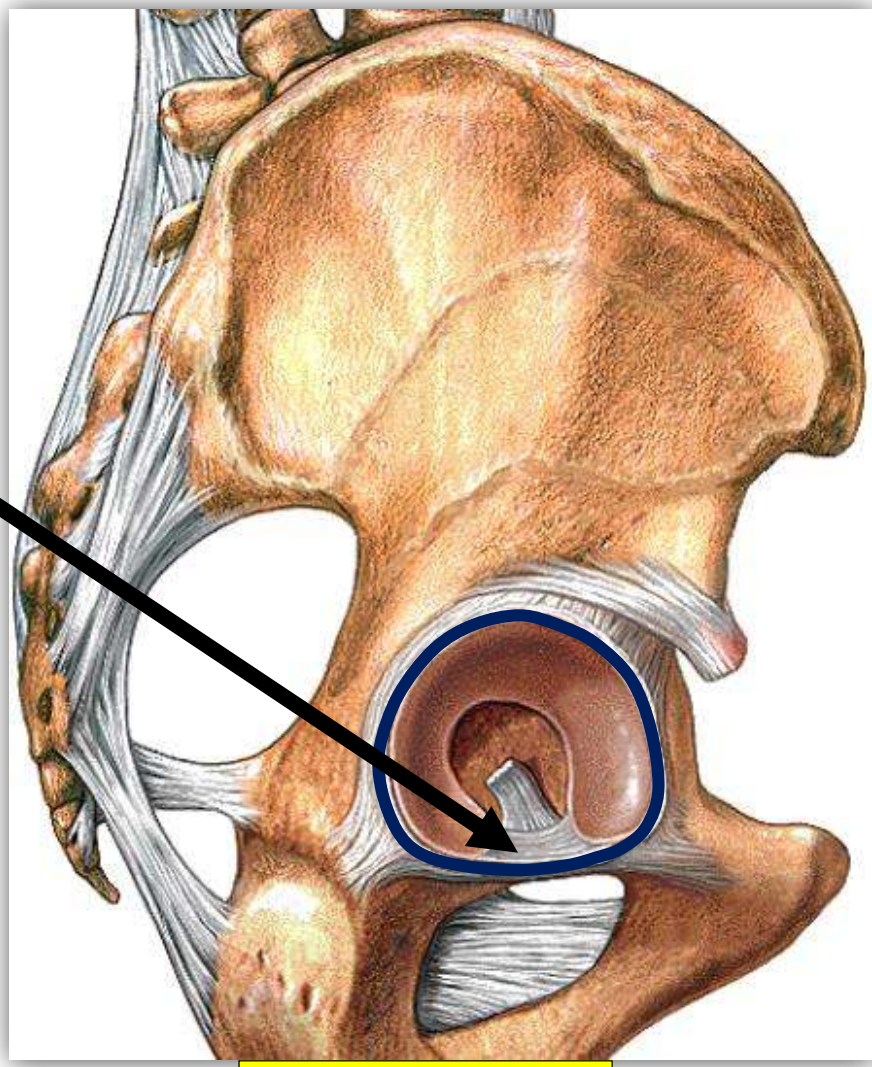
Labrum acétabulaire

- **Anneau fibro-cartilagineux**

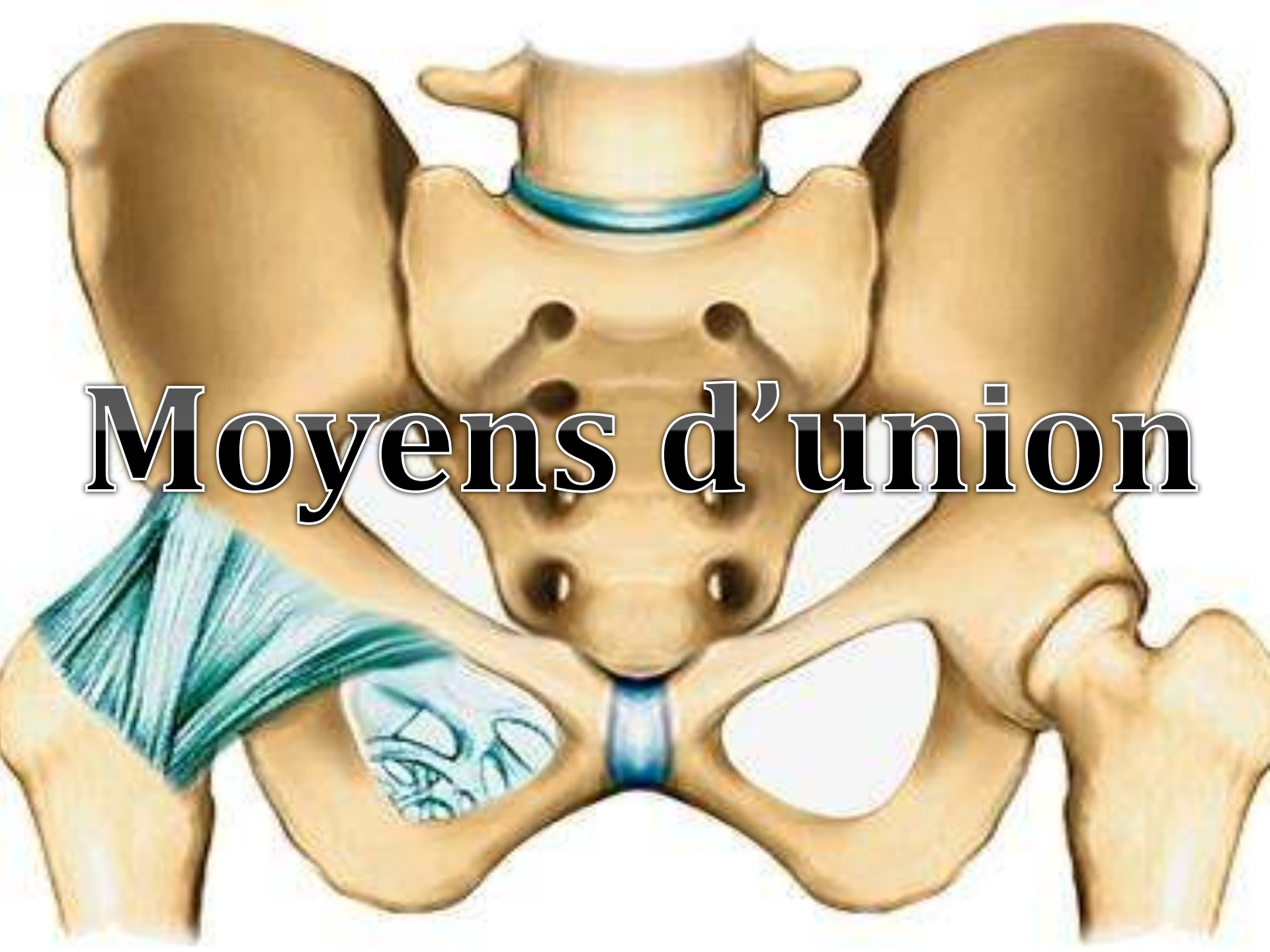
- Enroulé sur le **pourtour de l'acétabulum**

- Passe **en pont** au-dessus de **l'incisure ischio-pubienne**, formant le **ligament transverse de l'acétabulum**

- **Rôles**
  - **Il égalise le limbus acétabulaire**
  - **Augmente la profondeur** et l'**étendue de l'acétabulum**



Vue latérale

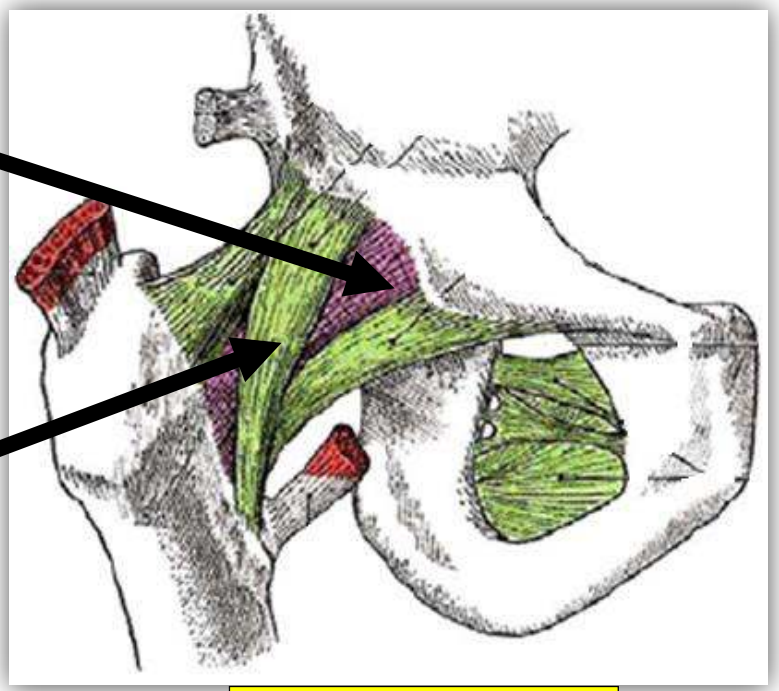


Moyens d’union

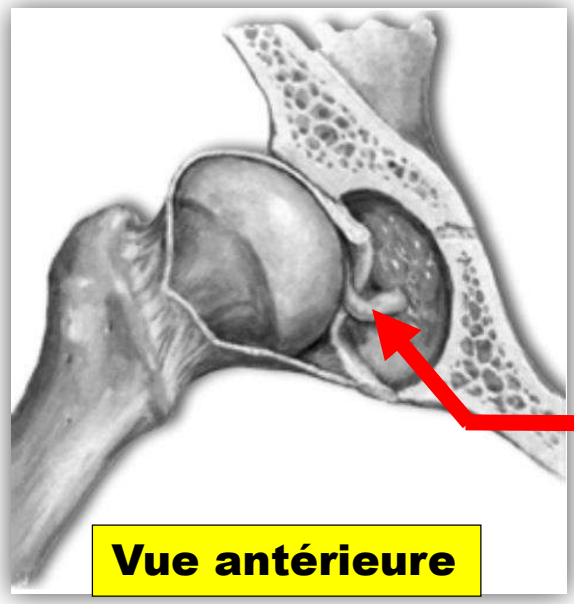
Maintiennent les surfaces articulaires en contact:

La capsule articulaire

Les ligaments Passifs et actifs qui Renforcent



Vue antérieure

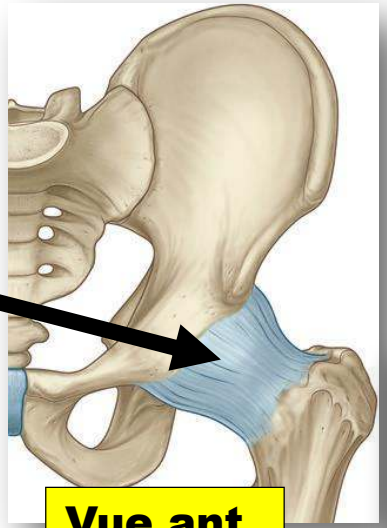


Vue antérieure

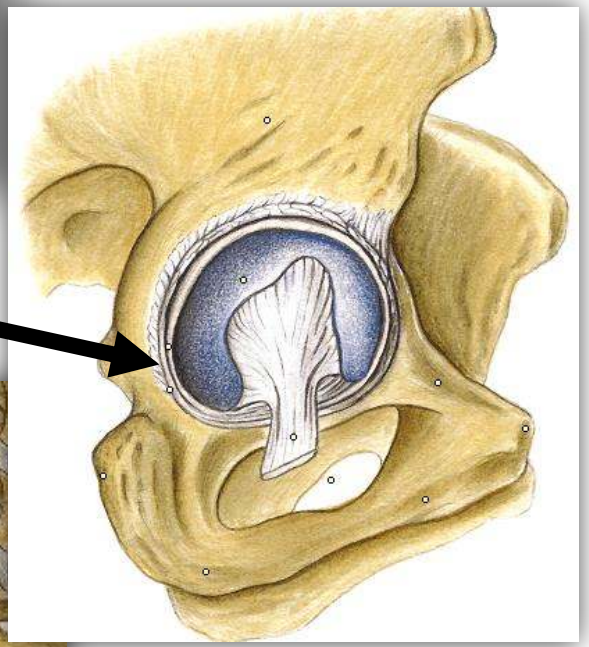
Le ligament de la tête fémorale (ligament rond) Indépendant de la capsule

Capsule articulaire

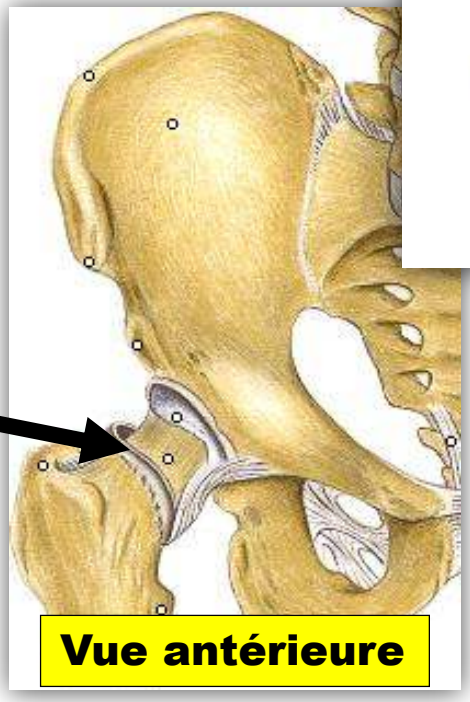
- **Manchon fibreux**
- **Insertion iliaque**
  sur le **limbus**
  et le **labrum acetabulaires**
- **Insertion fémorale:**
  sur la **ligne intertrochantérique**
  et **col fémoral**



Vue ant.



Vue latérale

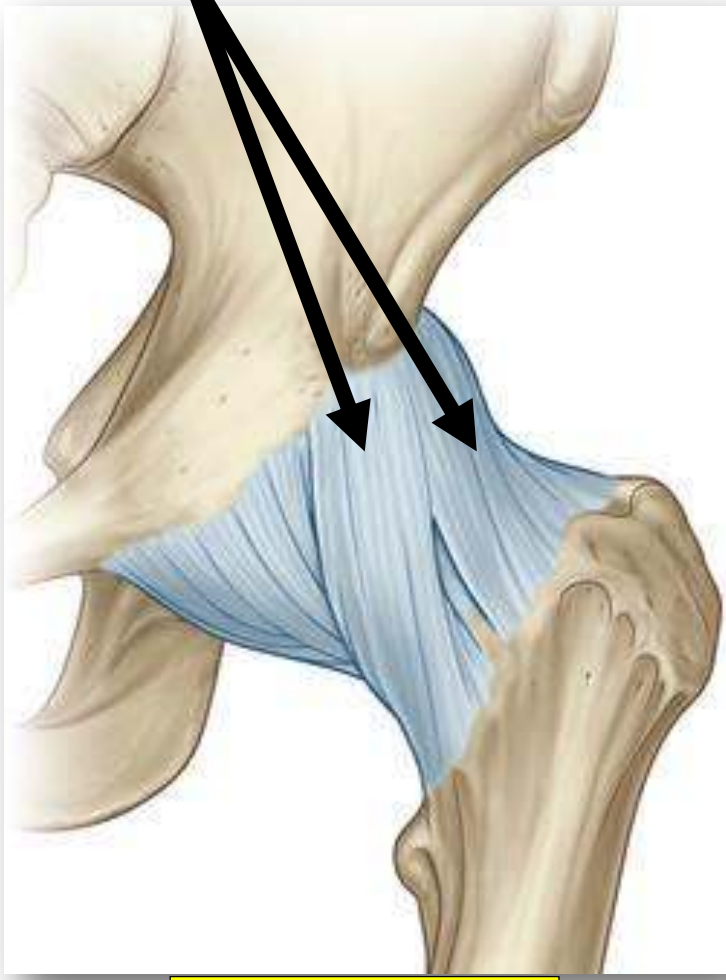


Vue antérieure

Ligaments passifs

Ligament ilio-fémoral
(le ligamentde Bertin)

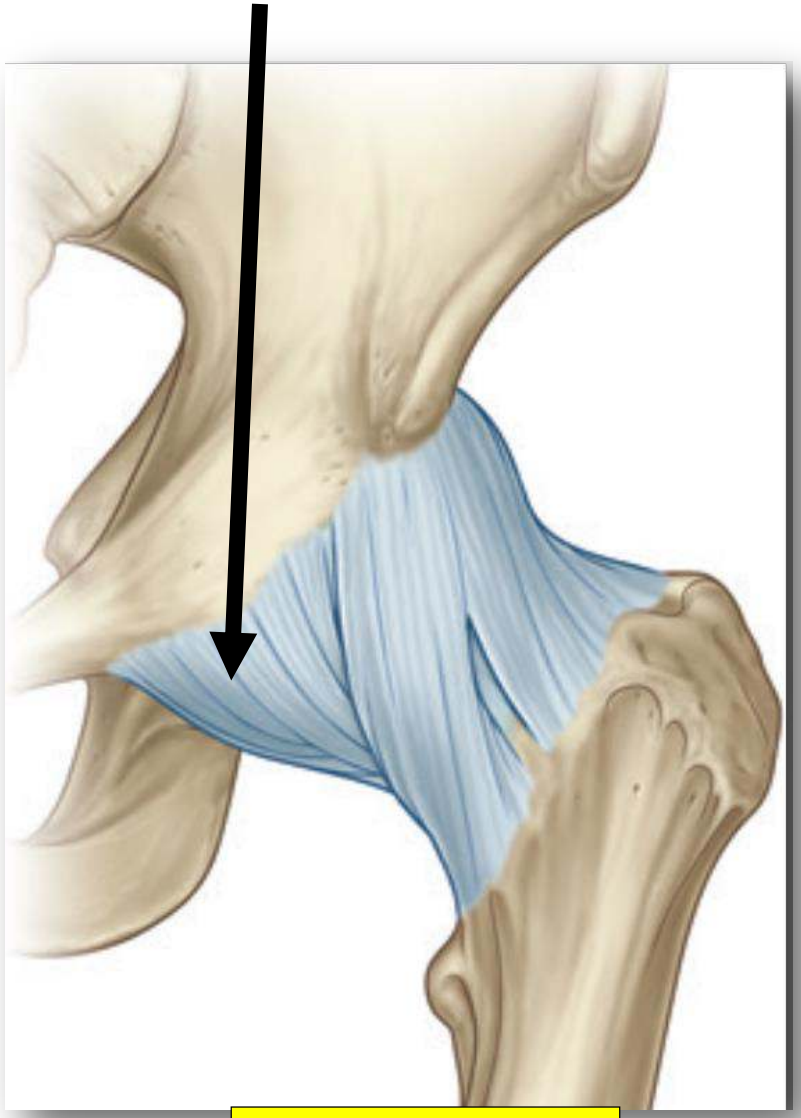
- **Antérieur**, **triangulaire** et **puissant**
- S'étend entre:
  l'**acétabulum** et la **ligne intertrochantérique**
- **2 faisceaux** distincts:
  - **Supérieur:**
    ilio-prétrochantérien
  - **Inférieur:**
    ilio-prétrochantinien



Vue antérieure

Ligament pubo-fémoral

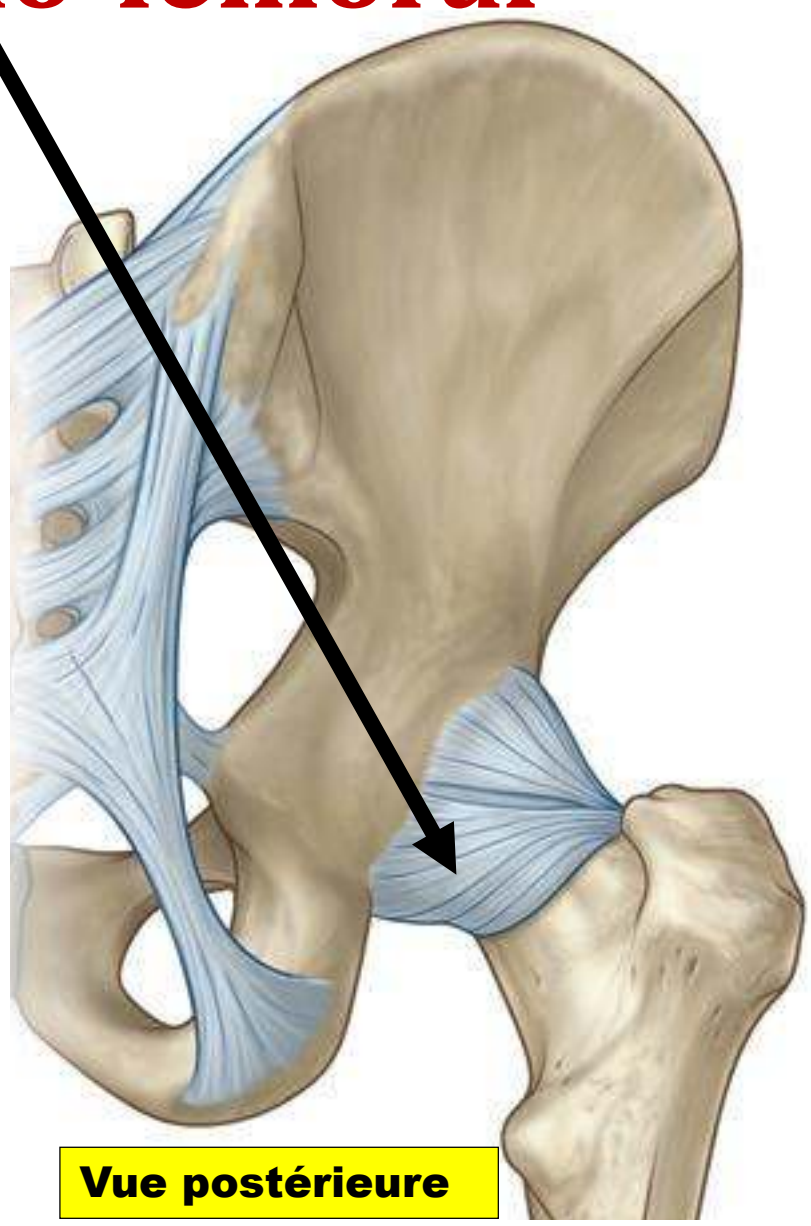
- Renforce la **partie antéro-inférieure** de la capsule articulaire
- S'étend entre l'**éminence ilio-pectinée** et la **dépression prétrochantinienne**



Vue antérieure

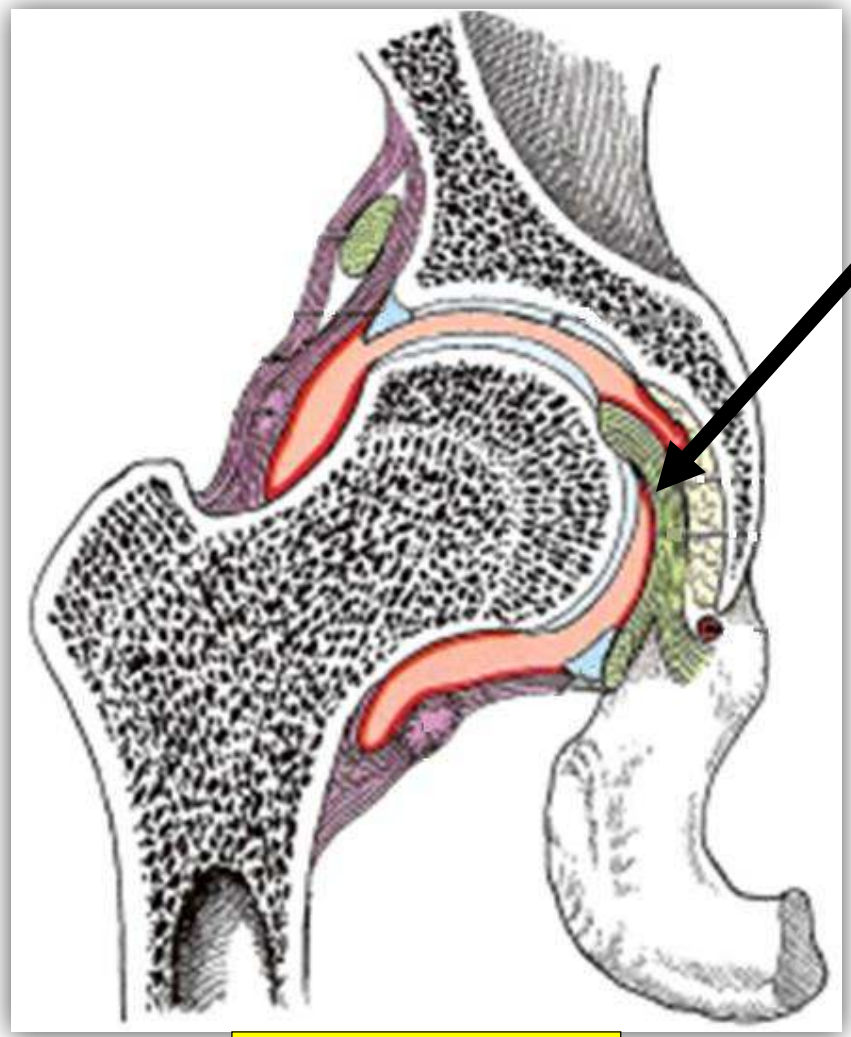
Ligament ischio-fémoral

- **Renforce en arrière** la capsule articulaire
- S'étend entre: **ischion** et **col fémoral**



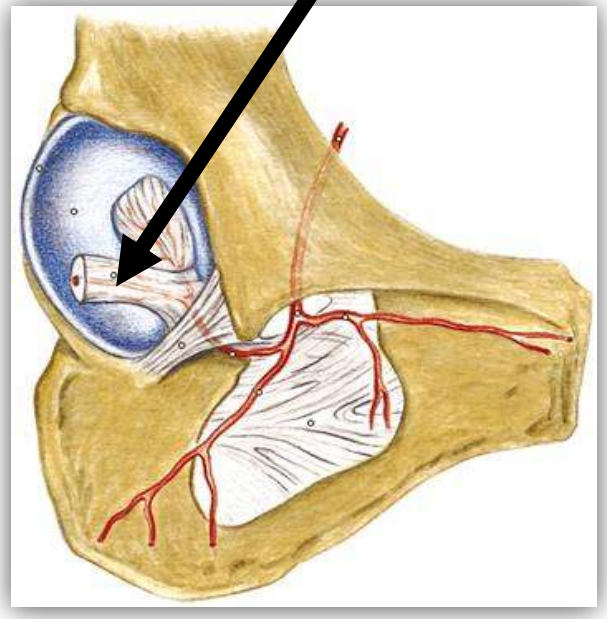
Vue postérieure

Ligament de la tête fémorale

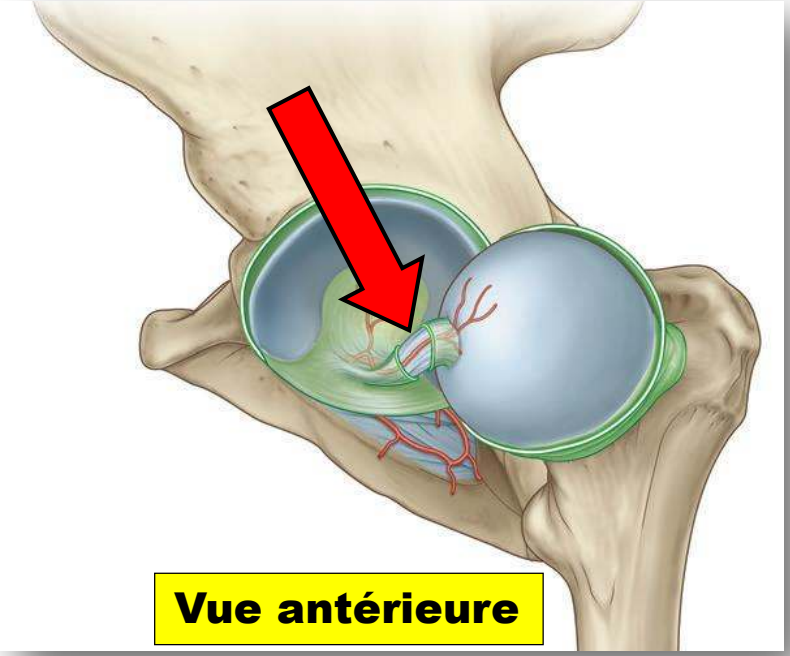


Coupe frontale

- **Intra-articulaire** et **extra-synovial**
- Centré par une **artère** qui irrigue la tête fémorale



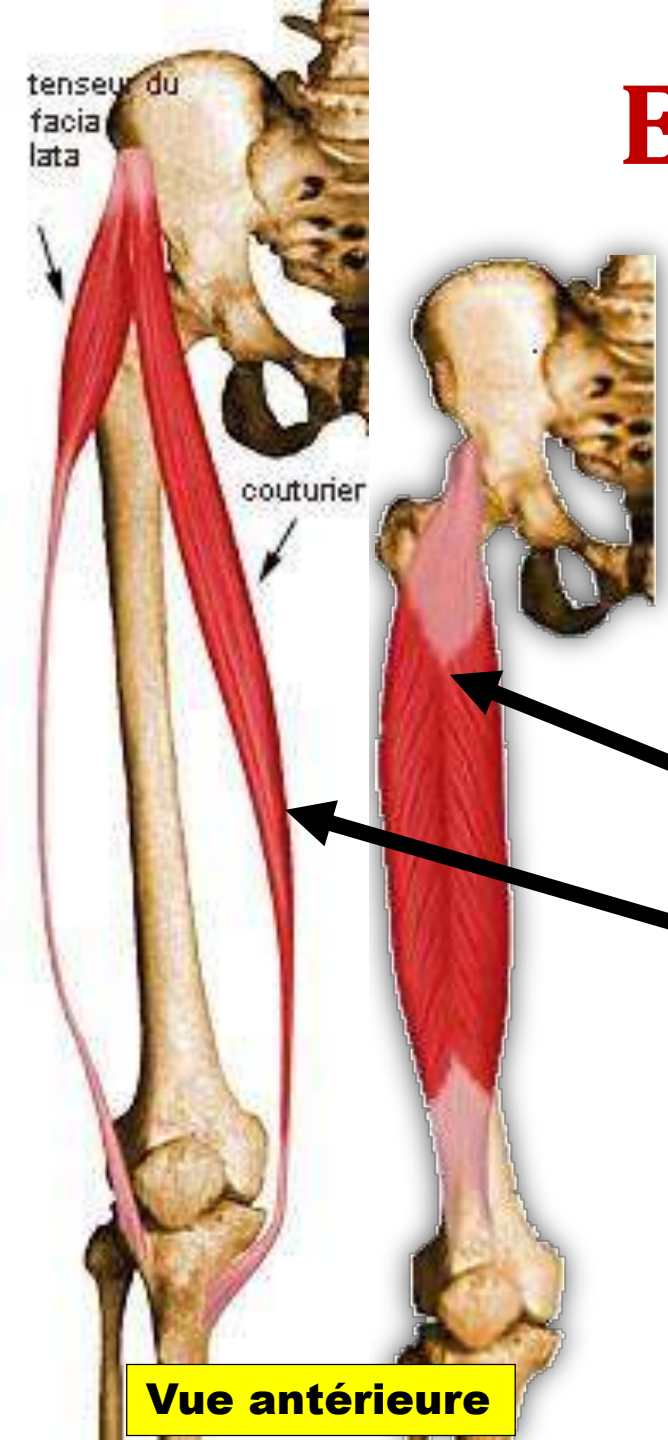
- S’étend de la **fovéa capitis**
- Se termine par **3 faisceaux**:
  - **Antérieur:**
    sur la **corne antérieure de la surface semi-lunaire**
  - **Moyen:**
    s’attache sur **ligament transverse**
  - **Postérieur:**
    sur l’ischion





Ligaments actifs (muscles péri-articulaires)

En avant



- Ilio-psoas
- Droit fémoral
- Sartorius

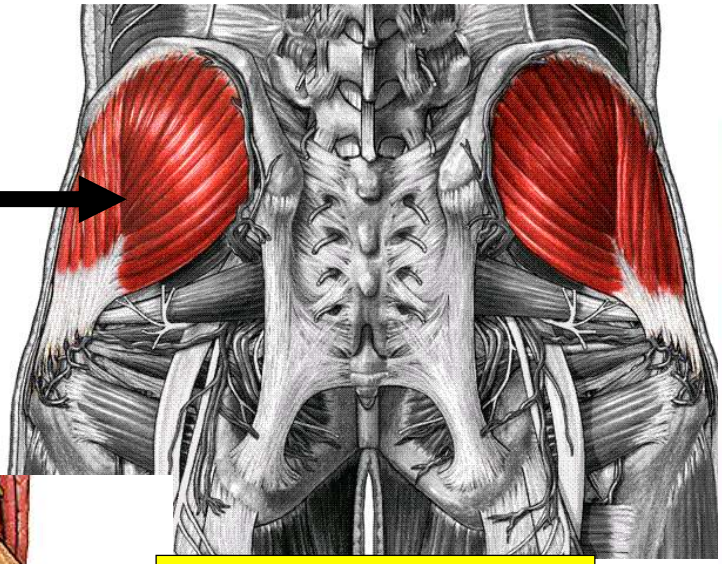


Vue antérieure

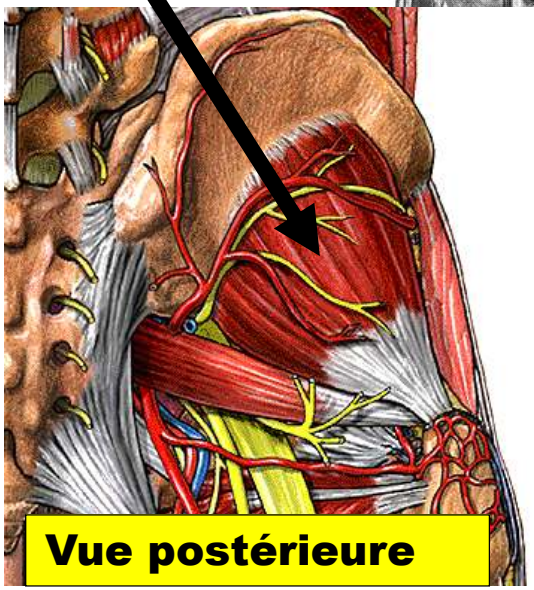
Vue antérieure

En dehors

- Le tenseur du fascia lata
- Moyen glutéal
- Petit glutéal



Vue postérieure

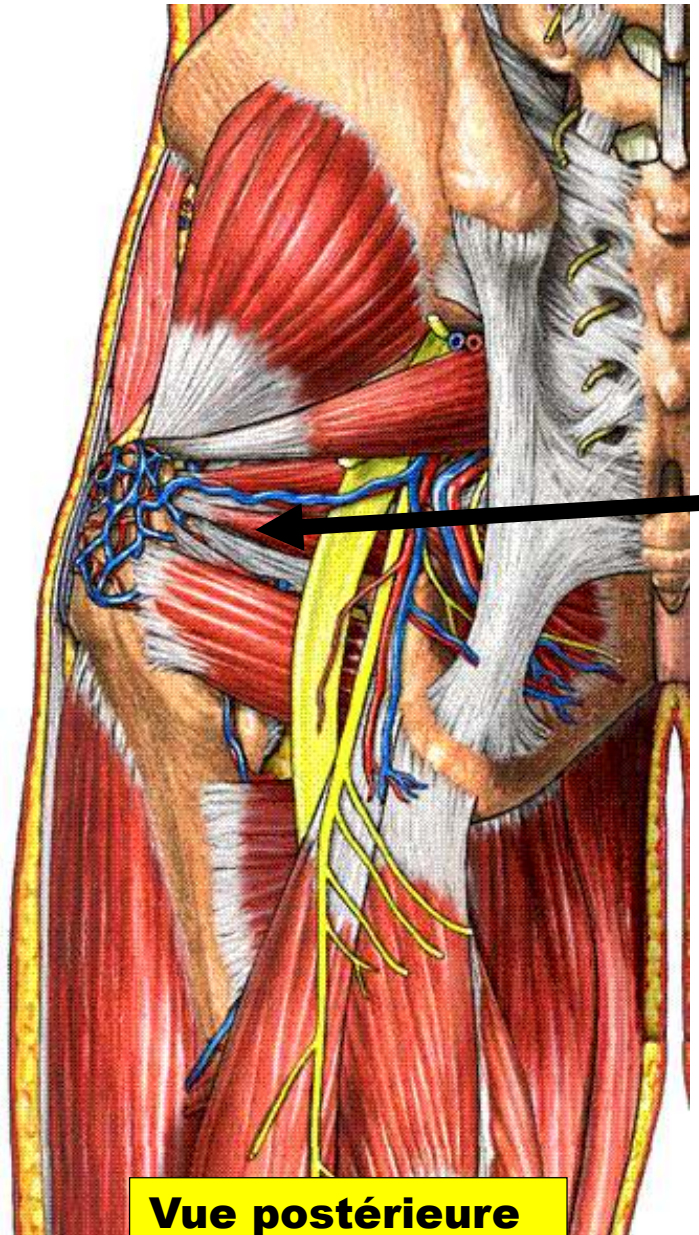


Vue postérieure

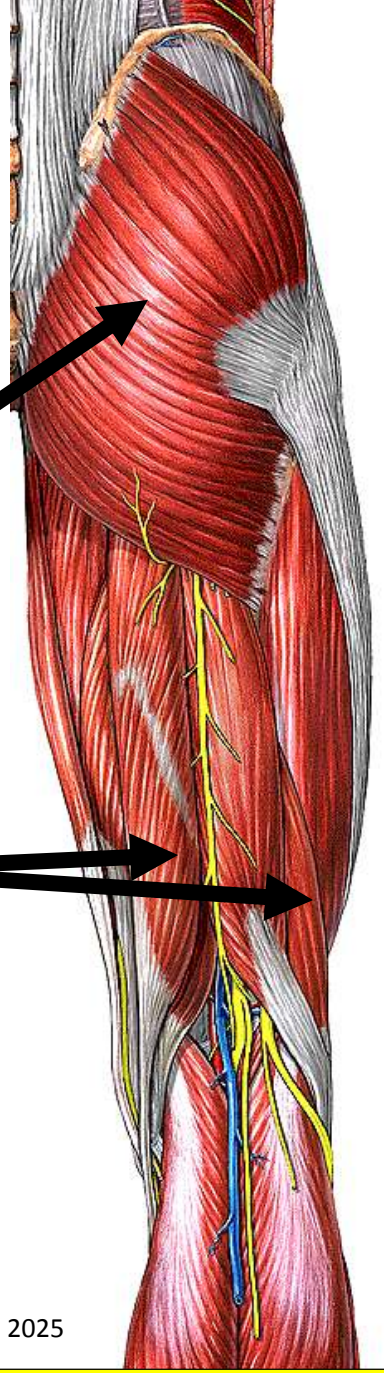


Vue antérieure

En arrière



- Grand glutéal
- Pelvi-trochantériens
- Ischio- jambiers



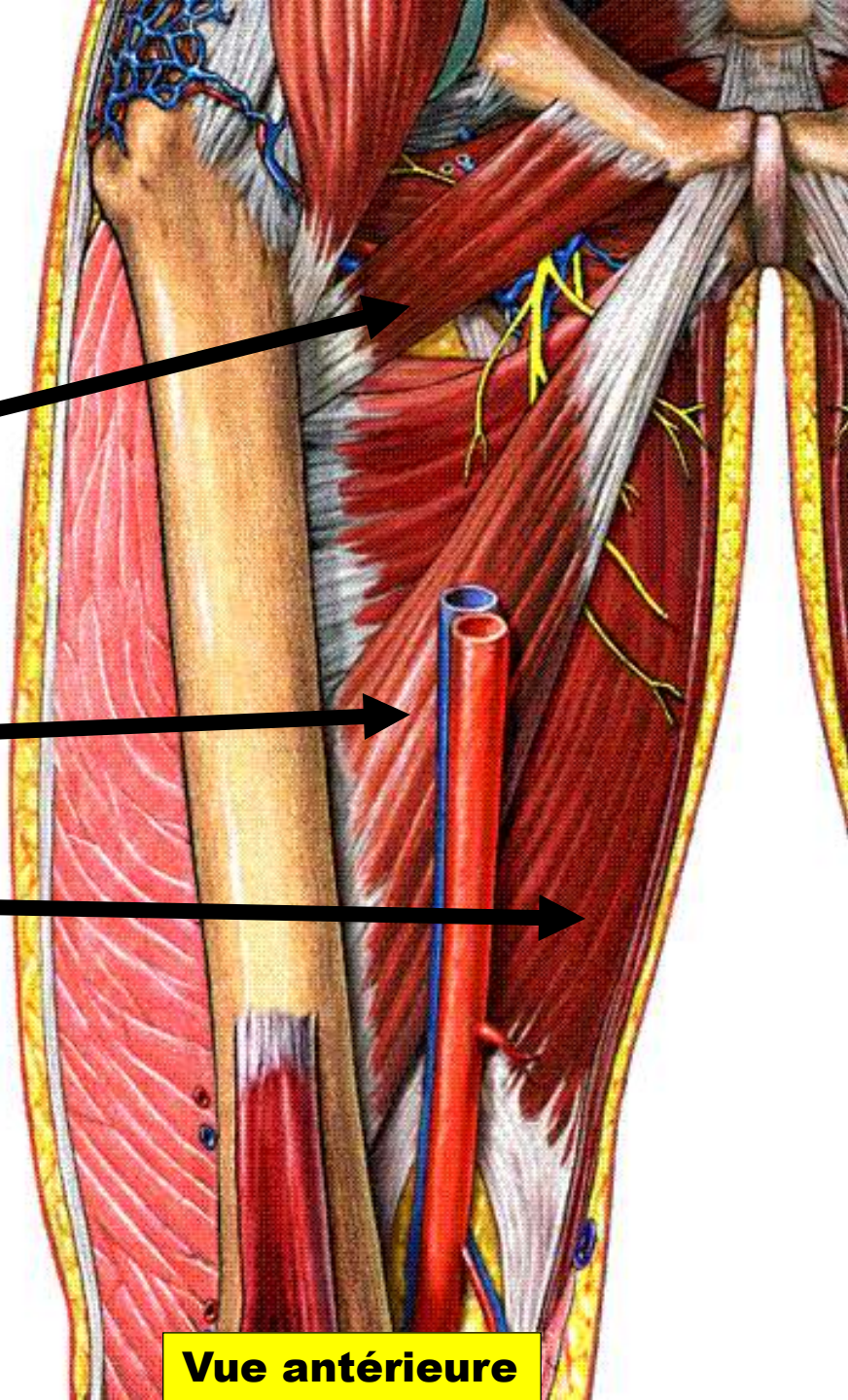
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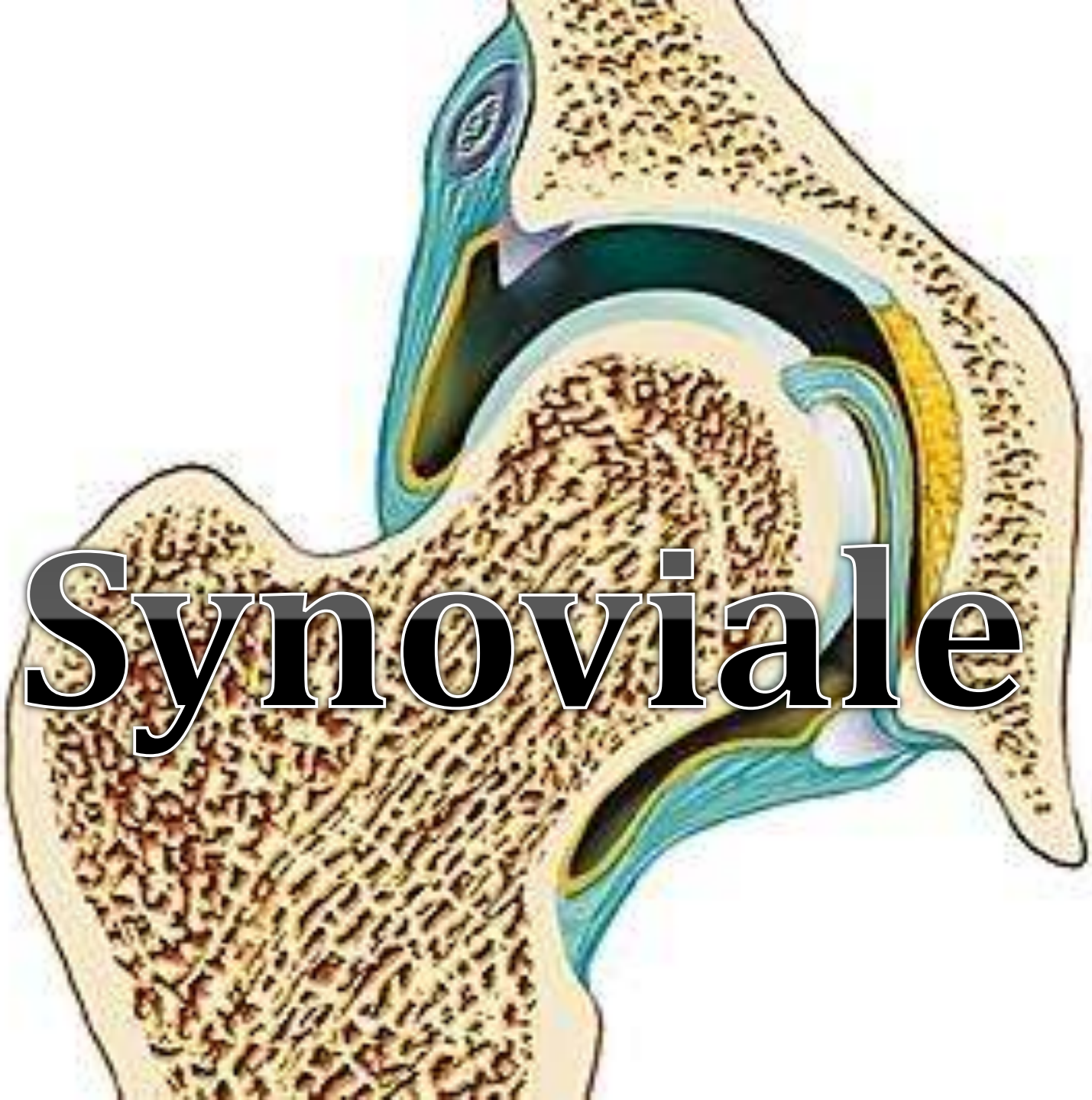
Vue postérieure

Vue postérieure

En dedans

- Pectiné
- Long adducteur
- Grand adducteur





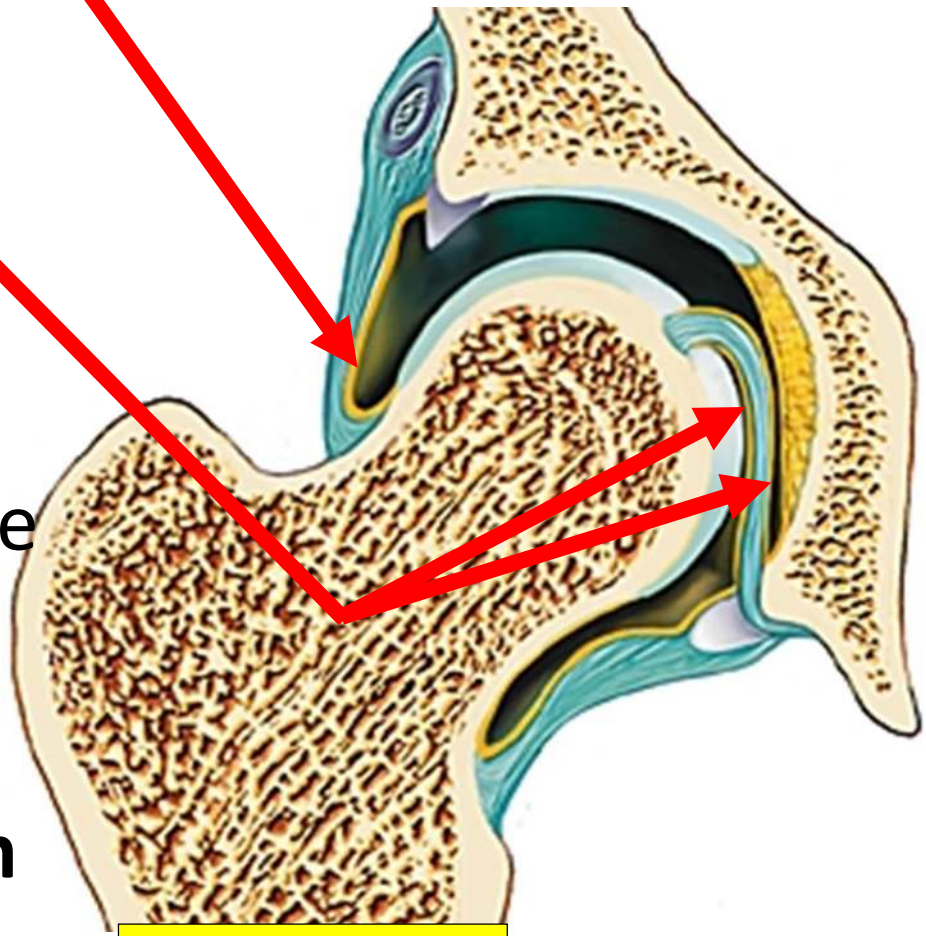
Synoviale

- La synoviale tapisse la **face profonde de la capsule**

- **Synoviale du ligament de la tête fémorale:**

  propre, **indépendante** de la synoviale articulaire

  S'attache **au pourtour de la fovéa capitis** et **fosse de l'acétabulum**

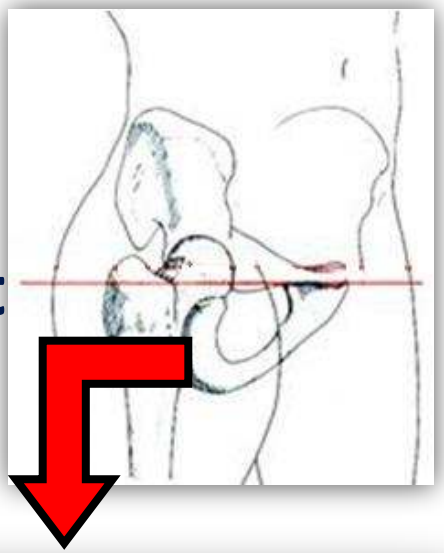


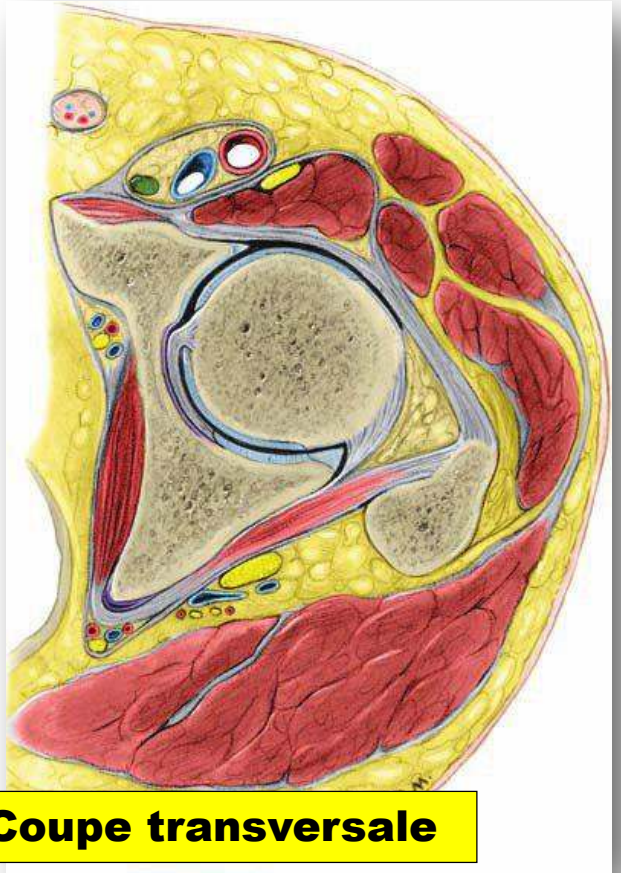
Coupe frontale



Physiologie articulaire

- **Articulation très stable** et **à grande mobilit**
- Sa stabilité est assurée par : la **capsule articulaire**, les **ligaments**, et les **muscles péri-articulaires**
- Permet **les mouvements**:
  - **Flexion-extension**
  - **Abduction-adduction**
  - **Rotation médiale-rotation latérale**





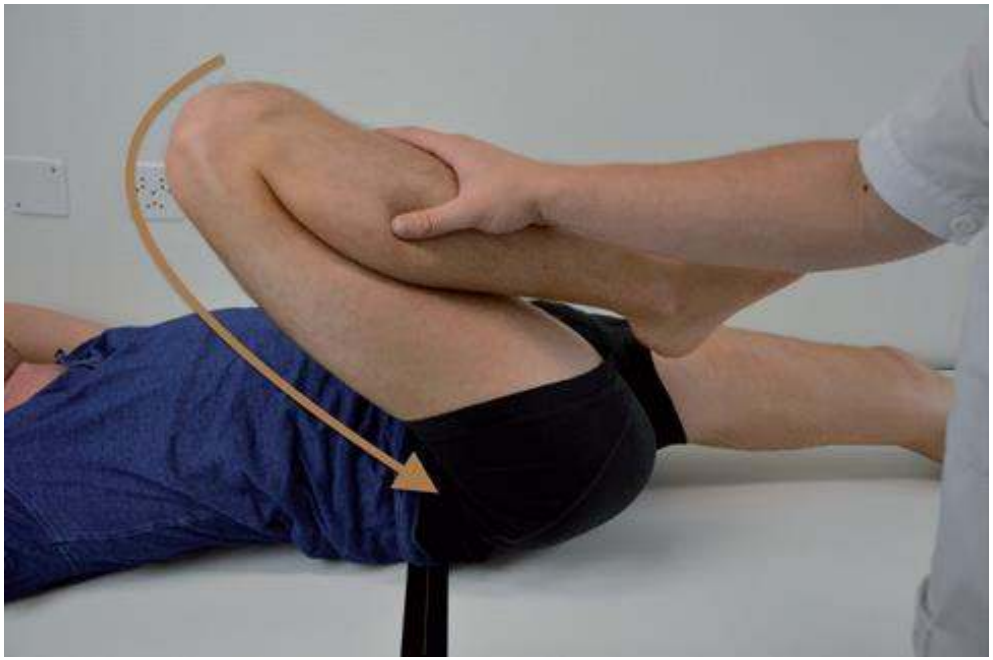
Coupe transversale

Flexion - Extension

La flexion et l'extension se font autour d'un **axe transversal** qui passe par **le centre de la tête fémorale**

Flexion

- **Rapproche la face antérieure de la cuisse** à **la paroi abdominale antérieure**
- **Amplitude:**
  - **120°** : genou fléchi
  - **90°** : genou étendu



Extension

- **Éloigne la face antérieure de la cuisse de la paroi abdominale antérieure**
- **Amplitude:**
  - **10°** : genou fléchi
  - **20°** : genou étendu

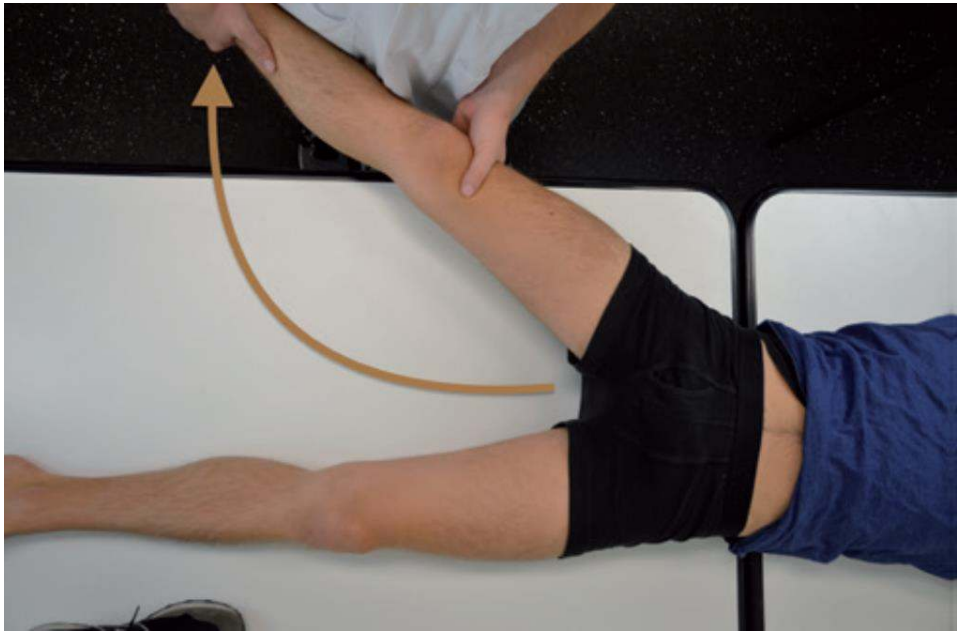


Abduction - Adduction

L’abduction et l’adduction se font autour d’un **axe antéro-postérieur** qui passe par **le centre de la tête fémorale.**

Abduction

- **Écarte la cuisse de la ligne médiane**
- **Amplitude:**
  - Habituelle: **30°**
  - Forcée: **45°** à **60°** (gymnastique…)



Adduction

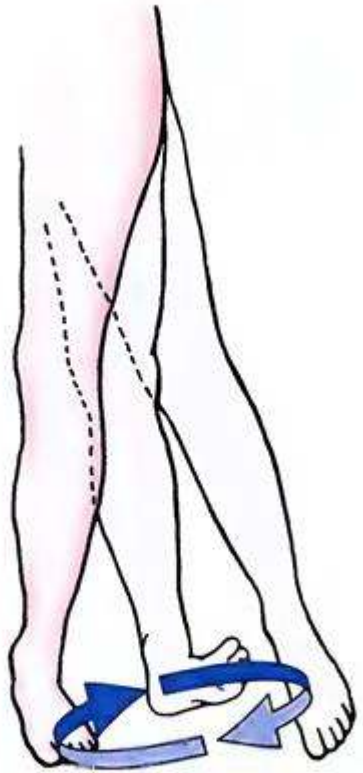
- **Rapproche: cuisse de la ligne médiane**
- **Amplitude: 30°** au maximum



Remarque: circumduction

- Résulte de **la succession**

  des **4 mouvements précédents**

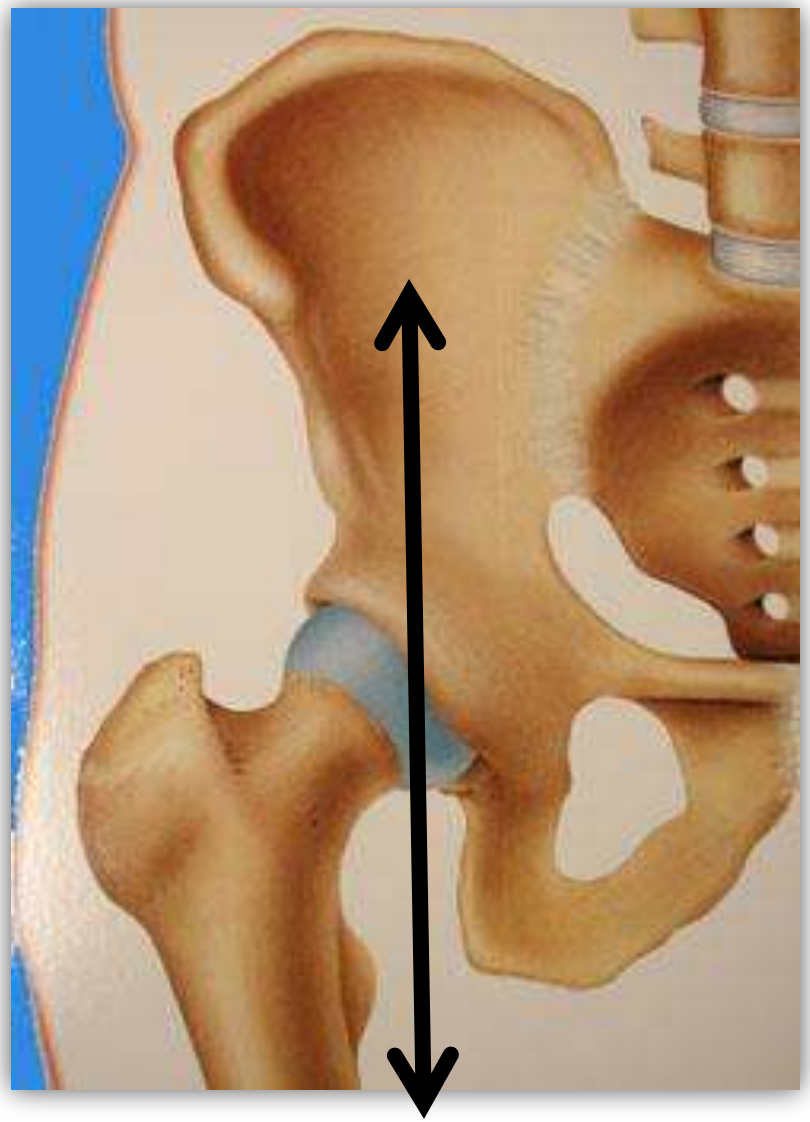


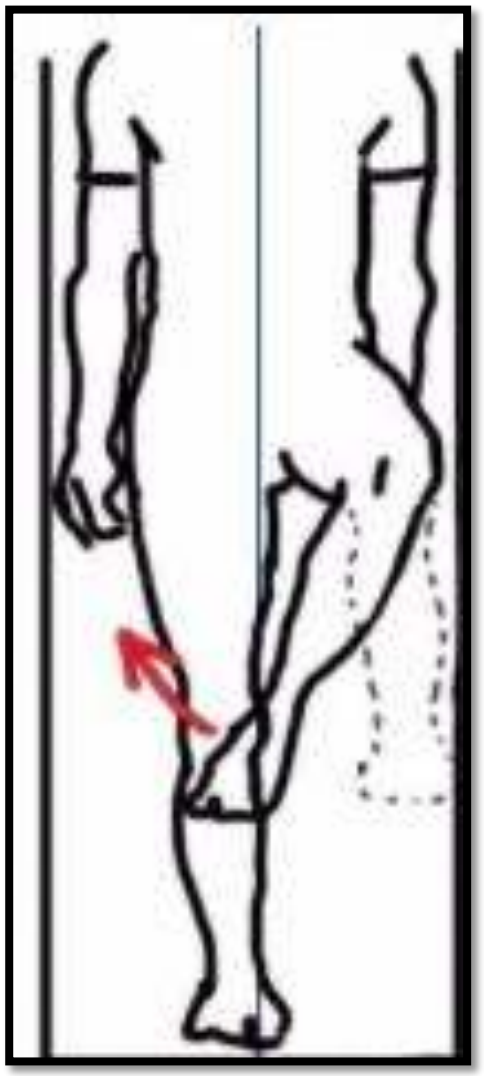
Circumduction

Circumduction (mouvement circulaire)
du membre inférieur au niveau de la hanche

Rotation médiale - Rotation latérale

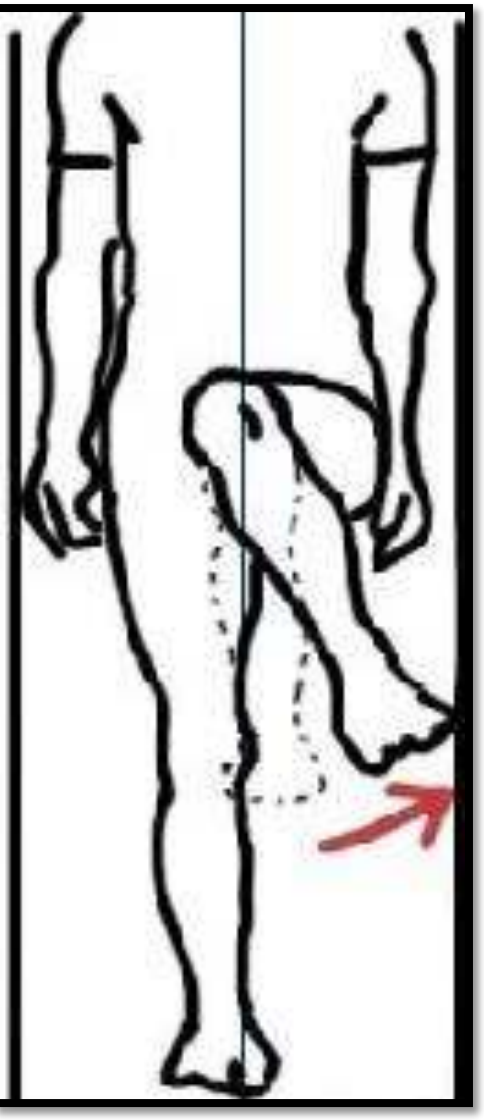
- Se font autour d'un **axe vertical** qui passe par **le centre de la tête fémorale**





Rotation latérale: 60°





Rotation médiale: 30°




Anatomie
clinique

Radio-anatomie

Hanche





hanche face et profil



arthroscanner



Pathologie

- **Arthroses**: **coxarthrose**
- **Arthrites**: inflammation ou infection grave (prothèses)
- **Fractures**
- **Luxations**
- **Luxations congénitales de la hanche (LCH)**

Arthroses: coxarthrose



Arthrites (prothèses)





Fractures du col du fémur





Luxations

(risque de rupture du ligament rond provoquant la nécrose de celle-ci)

Postérieure





Pubiennes et obturatrices (rares)

Luxation congénitale de la hanche (LCH)

- **Dépistée:**
  - À la naissance
  - Par la **manœuvre d'Ortolani** (cherche l'instabilité à la mobilisation des hanches)


Ortolani

Luxation congénitale de la hanche (LCH)



- **Traitement**:

  on lange

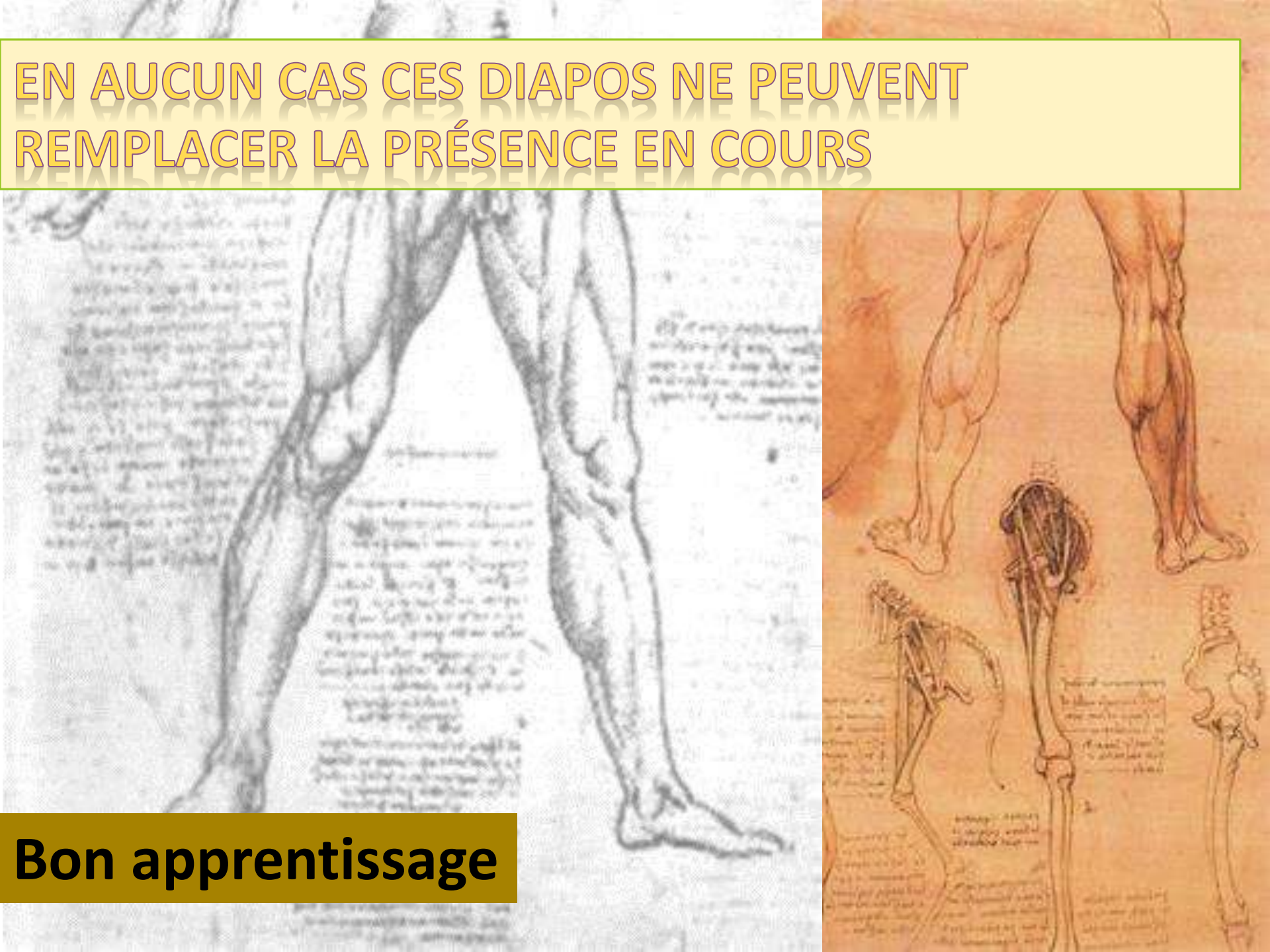
  le bébé

  en abduction

  forcée



EN AUCUN CAS CES DIAPOS NE PEUVENT REMPLACER LA PRÉSENCE EN COURS



Bon apprentissage